

Introduction to Algebra and Analysis

Part **Analysis part i**

B. McMaster

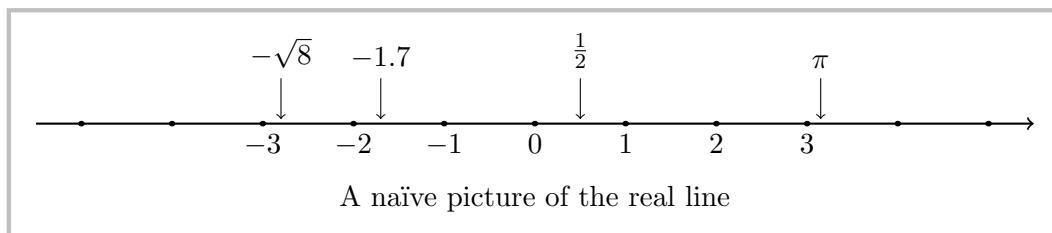
Contents

0.1	A checklist of things you probably know already	6
0.2	Modulus	7
0.2.1	Definition	7
0.2.2	Proposition	7
0.2.3	The triangle inequality	7
0.2.4	The reverse triangle inequality	7
0.3	Floor	8
0.3.1	Definition	8
1	Limit of a sequence	9
1.1	Sequences	9
1.1.1	Examples	10
1.2	Infinite decimals	11
1.3	Testing limits by the definition	12
1.3.1	Definition	12
1.3.2	Example	13
1.3.3	Example	13
1.3.4	Example	13
1.3.5	Remark	14
1.3.6	Example	14
1.3.7	Example	14
1.4	Combining sequences; the algebra of limits	14
1.4.1	Theorem	15
1.4.2	Example	16
2	Increasing and decreasing sequences	17
2.1	Introduction	17
2.2	Monotonic bounded sequences must converge	17
2.2.1	Definition	17
2.2.2	Example	18
2.2.3	Example	18
2.2.4	Definition	18
2.2.5	Lemma	18
2.2.6	Alert	19
2.2.7	Theorem	19
2.2.8	Example	19
2.2.9	Example	20

2.2.10	Theorem: limits across an inequality	20
2.2.11	Theorem: the “sandwich” or “squeeze”	20
2.2.12	Example	21
2.3	Recursively defined sequences	21
2.3.1	Example	22
2.3.2	Example	23
3	Sampling a sequence: subsequences	25
3.1	Subsequences	25
3.1.1	Definition	26
3.1.2	Examples	26
3.1.3	Theorem	27
3.1.4	Example	27
3.2	Bolzano-Weierstrass: the overcrowded interval	27
3.2.1	The Bolzano-Weierstrass theorem	28

Preliminaries: the basic rules of inequalities

You can choose to think of the real numbers as being all the possible decimals – finite and infinite, recurring and non-recurring, positive and negative and zero, whole numbers and fractions and surds¹ and non-surds such as π and e , and every possible combination of such objects. Equally well, you can choose to think of them as being (or being represented by) all the points that lie on a continuous unbroken straight line (the *real line*, the *real axis*) that stretches away endlessly in both directions. Somewhere on that line is a point marked 0 (zero) which separates the positives (on its right) from the negatives (on its left), and pacing out from zero at regular intervals in both directions lie the whole numbers (the *integers*) like distance markers along that endless road.



This is not, of course, a proper definition of what real numbers are. We are taking what is sometimes called a *naïve* view of the system of real numbers: not having sufficient time to construct it – to dig deeply enough into the logical foundations of mathematics to come up with a guarantee of its existence – we are instead seeking to highlight the common consensus on how real numbers behave, combine and *compare*. This consensus will already be enough to let us start explaining some basic ideas in analysis (and we shall say more about the finer structure of the real numbers in Chapter 2).

Nothing in section 0.1 is likely to strike the student reader as being much more than common sense, and nor should it at this stage of study. Nevertheless, it is all too easy to make mistakes in *comparisons between numbers – inequalities* – and it is consequently important to keep these apparently obvious rules in mind and to build up a good measure of confidence in their use, especially because so many arguments in analysis depend upon using inequalities. Sections 0.2 and 0.3 present a couple of useful operations on real numbers that are strongly connected with inequalities.

¹that is, non-rational numbers involving roots, such as $\sqrt{2}$, $\frac{\sqrt[3]{5}}{1 + \sqrt{2}}$, $\sqrt{10 - \sqrt[3]{2}}$.

0.1 A checklist of things you probably know already

- Each real number is either positive or zero or negative. “Non-negative” means positive or zero.
- $x > y$ and $y < x$ both mean $x - y$ is positive².
- $x \geq y$ and $y \leq x$ both mean $x - y$ is non-negative³.
- $x < y < z$ means both $x < y$ and $y < z$. Likewise for $>$, $\leq$, $\geq$.
- If $x < y$ and $y < z$, then $x < z$. Likewise for $>$, $\leq$, $\geq$.
- If $x \leq y$ and $y \leq x$, then $x = y$.
- If x and y are different real numbers then one of them is greater than the other, and is usually denoted⁴ by $\max\{x, y\}$.
- You can add a number to an inequality without damaging it:

$$x < y \implies x + a < y + a.$$
- You can add two inequalities:

$$(x < y \text{ and } a < b) \implies x + a < y + b.$$
- Notice how to use the symbol “ $\implies$ ” (pronounced *implies*): the last line is shorthand for “**if** $x < y$ **and** $a < b$ **then** $x + a < y + b$ ”.
- You can multiply an inequality **by a positive number** without damaging it:
 provided $a > 0$, we have $x < y \implies ax < ay$.
- If you multiply an inequality **by a negative number** it reverses it:
 provided $a < 0$, we have $x < y \implies ax > ay$.
- Provided the numbers involved are positive, you can take reciprocals across an inequality and it reverses it:

$$x < y \implies 1/x > 1/y \text{ provided that } x, y \text{ are positive.}$$
- Provided the numbers involved are positive, you can take square roots across an inequality and it preserves it:

$$x < y \implies \sqrt{x} < \sqrt{y} \text{ provided that } x, y \text{ are positive. Likewise for cube roots, fourth roots and so on.}$$
- “There are large integers:” that is, for any given real number x we can find an integer n so that $n > x$.

²— and are pronounced as x is greater/larger/bigger than y , y is less/smaller than x .

³— and are pronounced as x is greater than or equal to y , y is less than or equal to x .

⁴If $x = y$ then $\max\{x, y\}$ means x (or y , which is the same thing).

0.2 Modulus

0.2.1 Definition

If x is a real number, we define⁵ its *modulus* (also called its *absolute value*) as $|x|$ = the greater of x and $-x$. That is:

- If $x \geq 0$ then $|x| = x$;
- If $x < 0$ then $|x| = -x$.

Since the effect of modulus is to “throw away the minus from negative numbers”, the following should be obvious:

0.2.2 Proposition

For any real numbers x, y :

- $x \leq |x|, -x \leq |x|$,
- $|-x| = |x|$,
- $|xy| = |x| \cdot |y|$,
- $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$ provided that $y \neq 0$,
- $\sqrt{x^2} = |x|$.

0.2.3 The triangle inequality

For any real numbers x and y we have $|x + y| \leq |x| + |y|$.

Proof: Since $x \leq |x|$ and $y \leq |y|$, adding gives us $x + y \leq |x| + |y|$.

Exactly the same reasoning gives us $-x + (-y) = -(x + y) \leq |x| + |y|$.

Now $|x + y|$ is either $x + y$ or $-(x + y)$. So whichever one it is, it is $\leq |x| + |y|$.

Note: it is easy to extend this by induction⁶ to deal with any finite list of numbers, thus:

$$|x_1 + x_2 + x_3 + \dots + x_n| \leq |x_1| + |x_2| + |x_3| + \dots + |x_n|.$$

0.2.4 The reverse triangle inequality

For any real numbers x and y we have $||x| - |y|| \leq |x - y|$.

Proof: Use the triangle inequality on $x = (x - y) + y$ and we get $|x| \leq |x - y| + |y|$, from which $|x| - |y| \leq |x - y|$.

Interchange x and y , and we also get $|y| - |x| \leq |y - x| = |x - y|$.

Now $||x| - |y||$ is either $|x| - |y|$ or $|y| - |x|$. So whichever one it is, it is $\leq |x - y|$.

⁵More briefly: $|x| = \max\{x, -x\}$.

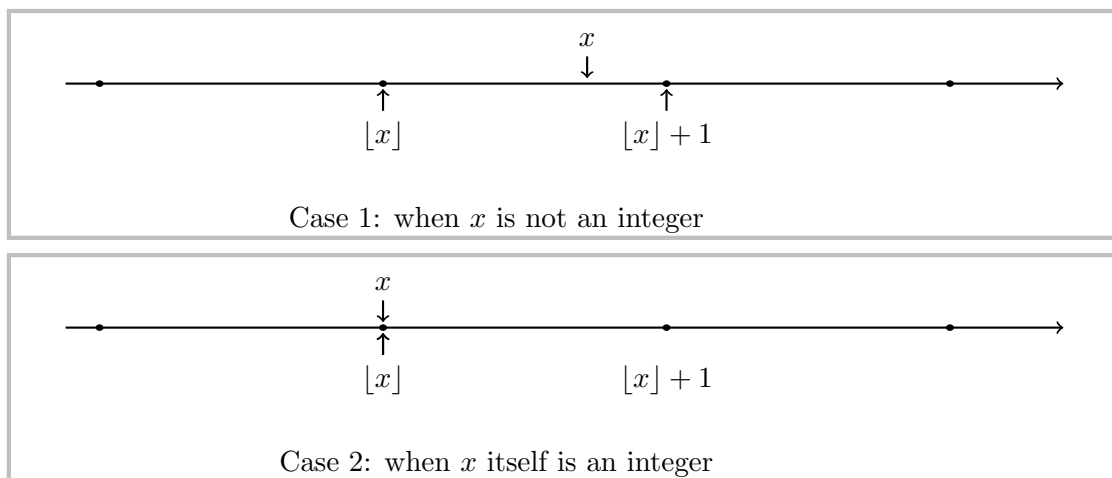
⁶We discuss this type of argument in detail later in the text.

0.3 Floor

0.3.1 Definition

When x is a real number, we define the *floor* of x (also called the *integer part* of x or, informally, x *rounded down to the nearest integer*) to be the largest integer that is $\leq x$. The usual notation for the floor of x is $\lfloor x \rfloor$, although some books write it as $[x]$.

If you choose to imagine the real numbers as being set out along the real line, with the integers – marked here by heavier dots – embedded into it at regular intervals, then the following diagram should help you to picture the relationship between x and $\lfloor x \rfloor$.



In both cases, the essential inequality connecting x and $\lfloor x \rfloor$ is

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$$

or, equivalently

$$x - 1 < \lfloor x \rfloor \leq x.$$

Chapter 1

Limit of a sequence

1.1 Sequences

A *sequence* is an unending list of real numbers in a particular order: a first one, a second, a third, and so on without end.

How to write a typical sequence:

$$(a_1, a_2, a_3, a_4, \dots, a_n, \dots)$$

$$(a_1, a_2, a_3, a_4, \dots)$$

$$(a_n)_{n \in \mathbb{N}}$$

$$(a_n)_{n \geq 1}$$

$$(a_n)$$

– and complete the description (if possible) by a formula for each a_n (the n^{th} term).

For instance, the sequence of all perfect squares could be written as:

$$(1, 4, 9, 16, \dots, n^2, \dots)$$

$$(1, 4, 9, 16, \dots)$$

$$(n^2)_{n \in \mathbb{N}}$$

$$(n^2)_{n \geq 1}$$

$$(n^2)$$

$$(a_1, a_2, a_3, a_4, \dots, a_n, \dots) \text{ where } a_n = n^2 \text{ for each positive integer } n$$

$$(a_1, a_2, a_3, a_4, \dots) \text{ where } a_n = n^2, \text{ each positive integer } n$$

$$(a_n)_{n \in \mathbb{N}} \text{ in which } a_n = n^2 \text{ for each } n$$

$$(a_n)_{n \geq 1} \text{ with } a_n = n^2 \text{ for each } n$$

$$(a_n), a_n = n^2 \text{ for each positive integer } n$$

- Be careful in notations like $(a_1, a_2, a_3, a_4, \dots, a_n, \dots)$ or $(a_1, a_2, a_3, a_4, \dots)$ not to leave out the final row of dots.
- Only use this style of notation if it is genuinely clear what the “pattern” is.
- Don’t leave off the enclosing brackets when writing a sequence.
- Sometimes it is convenient to start a sequence off at somewhere different from $n = 1$.

For example,

$$\left(\frac{n}{(n-1)(n-3)} \right)_{n \geq 4}, \quad (n!)_{n \geq 0}.$$

1.1.1 Examples

1. $(1, 3, 5, 7, 9, \dots) = (2n - 1)_{n \geq 1}$

2.
$$\left(\frac{3}{2}, \frac{3}{4}, \frac{9}{8}, \frac{15}{16}, \frac{33}{32}, \frac{63}{64}, \dots\right) = \left(1 + \frac{1}{2}, 1 - \frac{1}{2^2}, 1 + \frac{1}{2^3}, 1 - \frac{1}{2^4}, \dots\right)$$

$$= \left(\frac{2^n + (-1)^{n-1}}{2^n}\right)_{n \in \mathbb{N}}$$

3.
$$\left(5, \frac{1}{2}, 5, \frac{1}{4}, 5, \frac{1}{8}, 5, \frac{1}{16}, 5, \frac{1}{32}, \dots\right)$$

$$= (x_n) \text{ where } x_n = 5 \text{ if } n \text{ is odd but } x_n = 2^{-n/2} \text{ if } n \text{ is even.}$$

4. $(2, 3, 5, 7, 11, \dots) = (y_n)_{n \geq 1}$ where y_n is the n^{th} prime number. Notice how potentially misleading the first symbol was here! It could have meant several different sequences including, for example, “two, and then the odd integers excluding the perfect squares”. The second symbol was free from any such ambiguity.

5.
$$(1, -8, 27, -64, 125, -216, \dots) = (1, -8, 27, -64, \dots, (-1)^{n-1}n^3, \dots)$$

$$= ((-1)^{n-1}n^3)_{n \in \mathbb{N}}$$

Once again the first symbol might have been misunderstood, but the second and third left no room for confusion.

6.
$$\left(\frac{1}{1}, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \frac{1}{5}, -\frac{1}{6}, \frac{1}{7}, \dots\right) = \left(\frac{(-1)^{n-1}}{n}\right)_{n \geq 1}$$

7.
$$(1, \sqrt{2}, 3^{\frac{1}{3}}, 4^{\frac{1}{4}}, 5^{\frac{1}{5}}, \dots)$$

8.
$$\left((1+1)^1, \left(1+\frac{1}{2}\right)^2, \left(1+\frac{1}{3}\right)^3, \left(1+\frac{1}{4}\right)^4, \dots\right) = \left(\left(1+\frac{1}{n}\right)^n\right)_{n \in \mathbb{N}}$$

9.
$$\left(1, 1 + \frac{1}{2}, 1 + \frac{1}{2} + \frac{1}{3}, 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}, \dots\right)$$

10.

$$\left(1, 1 + \frac{1}{2}, 1 + \frac{1}{2} + \frac{1}{4}, 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}, \dots\right)$$

Notice that some sequences (but not all) seem to be settling towards an “equilibrium value”, a “steady state”, an “ultimate term” as we scan along the list. For instance, (2) above appears to be settling towards 1, and (6) towards 0; in contrast, (1) and (4) are so far showing no sign of settling, but are “exploding towards infinity” while (5) is doing some kind of cosmic splits by exploding towards infinity and minus infinity at the same time. In the case of (7) to (10) it is much less clear what is going to happen in the long run.

This idea of settling towards some limiting steady state is the most important property that a sequence can possess. Our task is to seek ways of deciding whether a given sequence ultimately settles or not, and if so, to what steady state it does “gravitate”.

1.2 Infinite decimals

In the online textbook you will find several case studies of situations where we generate better and better approximations to some “ideal” limiting number. Here we shall look in detail at just one simple one: approximating $\sqrt{2}$ by using more and more decimal places.

For each positive integer n let p_n stand for the decimal expansion, up to the n^{th} decimal digit, of $\sqrt{2}$. So, for a start:

$$\begin{aligned} p_1 &= 1.4 \\ p_2 &= 1.41 \\ p_3 &= 1.414 \\ p_4 &= 1.4142 \\ p_5 &= 1.41421 \\ &\vdots \end{aligned}$$

None of these numbers equals $\sqrt{2}$ exactly since each of them is rational and $\sqrt{2}$ is not rational. But they are approximations to it, and (broadly speaking) they provide better and better approximations as you work along the list: indeed, you could get “as close to $\sqrt{2}$ as you needed to be” just by going far enough.

To pin down the exact meaning of that last comment, look at the errors in the approximations, the differences $\sqrt{2} - p_n$:

$$\begin{aligned} \sqrt{2} - p_1 &= 0.01421356\dots < 0.1 &= 10^{-1} \\ \sqrt{2} - p_2 &= 0.00421356\dots < 0.01 &= 10^{-2} \\ \sqrt{2} - p_3 &= 0.00021356\dots < 0.001 &= 10^{-3} \\ &\vdots &&\vdots \\ \sqrt{2} - p_n &= < 0.00\dots 01 &= 10^{-n} \\ &\vdots &&\vdots \end{aligned}$$

We are under-informed as to the exact value of the stage- n error, but this is not going to matter: because, instead, we have an over-estimate of its size that is simple enough to work with. Look:

- If we are allowed a certain “tolerance of error”, that is, we’ve been asked to get some approximations whose actual errors are less than that tolerance, we can now easily see how to do it. Just find some positive integer N such that $(\frac{1}{10})^N$ is smaller than the permitted tolerance, and then p_N will be a good enough approximation because $\sqrt{2} - p_N < (\frac{1}{10})^N$ is smaller than the tolerance also.
- Continuing... not only is *that* p_N good enough, but all the later ones p_{N+1} , p_{N+2} , p_{N+3} , p_{N+4} ... will also be good enough in the same sense of “good”: they all have actual errors that are less than our allowed tolerance.

Incidentally, if we had approximated from above instead of from below, by opting for the list of numbers 1.5, 1.42, 1.415, 1.4143, 1.41422... then the differences $\sqrt{2} - p_n$ would have been negative. That would not have bothered us too much because the *size* of the error is usually more important than whether it is plus or minus. So we would have taken $p_n - \sqrt{2}$ as the error measurement in this case instead of $\sqrt{2} - p_n$, and the rest of the calculations would have worked out almost exactly the same.

To avoid worrying about whether our approximations are overestimates or underestimates just define the error to mean $|\sqrt{2} - p_n|$, so that error measurements are always counted as positive.

To compress our account of this string of improving approximations into a compact phrase, we use a standard symbol for the “tolerance of error”. The Greek letter ε is used. Thus, our precise and concise reason for declaring the list of numbers (p_n) to be a “good” approximation process for the square root of 2 is:

for each $\varepsilon > 0$ we can find a positive integer N such that
for every $n \geq N$, we get $|p_n - \sqrt{2}| < \varepsilon$.

1.3 Testing limits by the definition

1.3.1 Definition

1. A sequence $(x_n)_{n \in \mathbb{N}}$ is said to *converge* (or *tend*) to a *limit* L (where $L \in \mathbb{R}$) if:

for each $\varepsilon > 0$ there is some positive integer n_ε such that
 $|x_n - L| < \varepsilon$ for every $n \geq n_\varepsilon$.

2. Then we write $x_n \rightarrow L$ or $(x_n) \rightarrow L$ (as $n \rightarrow \infty$) or $\lim_{n \rightarrow \infty} x_n = L$. (The phrases “(as) $n \rightarrow \infty$ ” are often omitted.)
3. Not all sequences have limits. Those that do are called *convergent*, those that do not are called *divergent*.

Keep on thinking of x_n as the n^{th} item in a succession of approximations to L , and of ε as the tolerance for some intended application. In that sense the open interval $(L-\varepsilon, L+\varepsilon)$, which consists of exactly the numbers whose distances from L are smaller than ε , is where to find the “good” approximations – those whose errors are smaller than the current tolerance. Keep in mind that the physical distance between numbers x and y on the real line is $|x - y|$, so that the phrase $|x_n - L| < \varepsilon$ simply says “the distance between x_n and L is smaller than ε ”.

1.3.2 Example

Show that the harmonic sequence $\left(\frac{1}{n}\right)_{n \geq 1}$ converges to 0.

Solution

Given $\varepsilon > 0$, let n_ε be any integer larger than $\frac{1}{\varepsilon}$. Then for every integer $n \geq n_\varepsilon$ we have $n > \frac{1}{\varepsilon}$, therefore $\frac{1}{n} < \varepsilon$, that is, $\left|\frac{1}{n} - 0\right| < \varepsilon$. By definition, $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

1.3.3 Example

Put $x_n = 4 - 3n^{-2}$ for each positive integer n . Show that (x_n) is a convergent sequence, and that its limit is 4.

Solution

Given $\varepsilon > 0$, let n_ε be any integer larger than $\sqrt{\frac{3}{\varepsilon}}$. Then for every integer $n \geq n_\varepsilon$ we have $n > \sqrt{\frac{3}{\varepsilon}}$ and therefore $n^2 > \frac{3}{\varepsilon}$ and therefore $\frac{1}{n^2} < \frac{\varepsilon}{3}$ and therefore $\frac{3}{n^2} < \varepsilon$, that is,

$$|x_n - 4| = |4 - 3n^{-2} - 4| = \frac{3}{n^2} < \varepsilon.$$

By definition, the sequence (x_n) converges to 4.

1.3.4 Example

Show that

$$\lim_{n \rightarrow \infty} \frac{n(3n-1)}{n^2+1} = 3.$$

Solution

Given $\varepsilon > 0$, let n_ε be any integer larger than $\frac{4}{\varepsilon}$. Then for every integer $n \geq n_\varepsilon$ we have $n > \frac{4}{\varepsilon}$ and therefore $\frac{n}{4} > \frac{1}{\varepsilon}$ and therefore $\frac{4}{n} < \varepsilon$. But

$$|x_n - 3| = \left| \frac{n(3n-1)}{n^2+1} - 3 \right| = \frac{n+3}{n^2+1} < \frac{n+3}{n^2} \leq \frac{n+3n}{n^2} = \frac{4n}{n^2} = \frac{4}{n}$$

so also $|x_n - 3| < \varepsilon$. By definition, $x_n \rightarrow 3$.

1.3.5 Remark

How a sequence converges (or not) is not influenced in any way by the first few terms... nor, indeed, by the first trillion terms. For imagine that we take a convergent sequence (a_n) with limit L , and alter the first trillion ($= 10^{12}$) terms in some fashion. Given positive ε , we can find n_ε so that $n \geq n_\varepsilon$ makes the error terms $|a_n - L|$ in the original sequence less than ε . If it happens that n_ε is more than a trillion, this remains true for the modified sequence (since the modifications only affected the early terms). Yet if n_ε is a trillion or less, we see that $n \geq 10^{12}$ forces the n^{th} stage errors in both the modified and the unmodified sequences to be smaller than ε once again. In both cases, the limit has not been affected.

This allows us, when exploring the limit of a sequence, to ignore the first few (or the first many! but never *infinitely* many) terms if it simplifies our argument. Here is an illustration:

1.3.6 Example

Show from the definition that $\frac{4n}{2n-7} \rightarrow 2$.

Solution

If $n \geq 7$ then we get $-7 \geq -n$, $2n - 7 \geq 2n - n = n$, $\frac{1}{2n-7} \leq \frac{1}{n}$, $\frac{14}{2n-7} \leq \frac{14}{n}$. Now, given $\varepsilon > 0$, choose a positive integer $n_0 > \frac{14}{\varepsilon}$. For any $n \geq n_0$ we have $n > \frac{14}{\varepsilon}$ and therefore $\frac{1}{n} < \frac{\varepsilon}{14}$, so:

$$\left| \frac{4n}{2n-7} - 2 \right| = \left| \frac{14}{2n-7} \right| = \frac{14}{2n-7} \leq \frac{14}{n} < 14 \frac{\varepsilon}{14} = \varepsilon.$$

We ignored the first six terms but that has no effect on limits. So the result is proved.

1.3.7 Example

Show that each constant sequence converges.

Solution

Consider a sequence (x_n) in which *every* x_n is the same number: a sequence of the form $(c, c, c, c, c, \dots)$. We show that the limit of (x_n) is also c . (In other notation, $\lim_{n \rightarrow \infty} c = c$.)

Given $\varepsilon > 0$, let us choose $n_\varepsilon = 1$. Then for every $n \geq n_\varepsilon$ we get $|x_n - c| = |c - c| = 0 < \varepsilon$ and the proof is complete.

1.4 Combining sequences; the algebra of limits

This section offers the first of a number of quicker ways to establish limits for relatively complicated sequences. The basic idea is: if you can see how to build up a sequence from simpler ones whose limits you already know, then you can write down the limit of the complicated one just by combining the limits of its simpler “components”.

1.4.1 Theorem

Suppose that (a_n) and (b_n) are convergent sequences, with limits A and B respectively. Then:

1. $a_n + b_n \rightarrow A + B$,
2. $a_n - b_n \rightarrow A - B$,
3. $a_n b_n \rightarrow AB$,
4. For each constant k , $ka_n \rightarrow kA$,
5. $|a_n| \rightarrow |A|$,
6. Provided that $B \neq 0$ and that no b_n is zero, also $\frac{a_n}{b_n} \rightarrow \frac{A}{B}$.

REMARK: There are several other ways of expressing this, including:

1. $\lim(a_n + b_n) = \lim a_n + \lim b_n$,
2. $\lim(a_n - b_n) = \lim a_n - \lim b_n$,
3. $\lim(a_n b_n) = \lim a_n \lim b_n$,
4. For each constant k , $\lim(ka_n) = k \lim a_n$,
5. $\lim |a_n| = |\lim a_n|$,
6. Provided that no division by zero occurs, $\lim \frac{a_n}{b_n} = \frac{\lim a_n}{\lim b_n}$.

(The entire result can even be turned into English, thus: taking limits of convergent sequences is compatible with addition, with subtraction, with multiplication, with “scaling” (that is, multiplying by constants), with taking modulus, and with division provided always that no illegal division by zero is attempted.)

Proof of part (1)

Let $\varepsilon > 0$ be given. Then $\varepsilon/2 > 0$ also! Since we know that (a_n) converges to A , there is some positive integer n_1 such that

$$n \geq n_1 \text{ forces } |a_n - A| < \frac{1}{2}\varepsilon.$$

For similar reasons there is another positive integer n_2 such that

$$n \geq n_2 \text{ forces } |b_n - B| < \frac{1}{2}\varepsilon.$$

Write $n_0 = \max\{n_1, n_2\}$, and notice that whenever $n \geq n_0$ we get both the displayed lines working at once. Therefore, for every $n \geq n_0$:

$$|a_n + b_n - (A + B)| = |(a_n - A) + (b_n - B)| \leq |a_n - A| + |b_n - B| < \frac{1}{2}\varepsilon + \frac{1}{2}\varepsilon = \varepsilon.$$

The proof of (1) is complete.

Proof of part (4)

In the special case $k = 0$ we know this already: because then (ka_n) is a constant sequence of zeroes, kA is exactly zero, and an earlier example told us that $\lim 0 = 0$. So for the rest of the proof we can assume that $k \neq 0$.

Given any $\varepsilon > 0$, notice that $\frac{\varepsilon}{|k|} > 0$ also. Since $a_n \rightarrow A$, there is a positive integer n_0 such that

$$n \geq n_0 \text{ forces } |a_n - A| < \frac{\varepsilon}{|k|}.$$

Then also

$$n \geq n_0 \text{ forces } |ka_n - kA| = |k(a_n - A)| = |k||a_n - A| < |k| \frac{\varepsilon}{|k|} = \varepsilon$$

which is the definition of $ka_n \rightarrow kA$.

Moving towards the *use* of this theorem, notice that (by part (3))

$$\lim \left(\frac{1}{n^2} \right) = \lim \left(\frac{1}{n} \frac{1}{n} \right) = \lim \left(\frac{1}{n} \right) \lim \left(\frac{1}{n} \right) = 0.0 = 0$$

and, consequently, that

$$\lim \left(\frac{1}{n^3} \right) = \lim \left(\frac{1}{n^2} \frac{1}{n} \right) = \lim \left(\frac{1}{n^2} \right) \lim \left(\frac{1}{n} \right) = 0.0 = 0$$

and so on. Consequently $\frac{1}{n^k} \rightarrow 0$ for each positive integer k . (It is easy to prove this formally by induction if you wish to try it.) This observation, combined with pieces of our main theorem, allows us to deal with a large number of limit problems quickly and painlessly. For instance:

1.4.2 Example

Establish the convergence of the sequence (a_n) described by

$$a_n = \frac{15n^2 + n + 1}{5n^2 - n - 2}.$$

Solution

Begin by dividing the top and bottom lines of the fraction formula by the largest power of n appearing (n^2):

$$a_n = \frac{15n^2 + n + 1}{5n^2 - n - 2} = \frac{15 + n^{-1} + n^{-2}}{5 - n^{-1} - 2n^{-2}} \rightarrow \frac{15 + 0 + 0}{5 - 0 - 2(0)} = \frac{15}{5} = 3.$$

Finished already! Notice how many aspects of the key theorem were used in that last line: part (6) to let us deal with the top and bottom lines separately, parts (1) and (2) to add/subtract the separate parts of each line, part (4) to see that $\lim 2n^{-2} = 2 \lim n^{-2}$. It is not necessary to write out all of these moves, but be aware of them.

Chapter 2

Increasing and decreasing sequences

2.1 Introduction

If a sequence converges to a limit, then it is perfectly possible that it does not do so *steadily*. For instance, the sequence $(\frac{1}{1}, \frac{1}{3}, \frac{1}{2}, \frac{1}{4}, \frac{1}{3}, \frac{1}{5}, \frac{1}{4}, \frac{1}{6}, \dots)$ tends to 0, but in a “two steps forward, one step back” fashion: as you scan along the list of terms, you find that you are moving *away* from 0 half the time. On the other hand, the sequence $(\frac{1}{1}, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \frac{1}{5}, -\frac{1}{6}, \frac{1}{7}, -\frac{1}{8}, \dots)$ converges to zero “in both directions at once”, alternately over- and underestimating the eventual limit. However, many important sequences move *steadily* in one direction (up or down) and, for those that do, it is often easier to determine whether they converge or not. This chapter will focus on sequences of this kind.

2.2 Monotonic bounded sequences must converge

2.2.1 Definition

A sequence $(x_n)_{n \geq 1}$ is called

- an *increasing* sequence if $x_n \leq x_{n+1}$ for every n ;
- a *decreasing* sequence if $x_n \geq x_{n+1}$ for every n .
- a *monotonic* (or *monotone*) sequence if it is either increasing or decreasing.

These are the sequences that “move steadily in one direction – up or down” as you scan along the list of terms. There are some very obvious examples, such as the decreasing sequences $(\frac{1}{n})$ and $(\frac{1}{n^2})$ and $(\frac{1}{n^3})$ and $(-n)$ and $(-\sqrt{n})$, and the increasing sequences (n) and (n^2) and $(1 - \frac{1}{n})$. For less transparent examples, it is often useful to calculate the difference $x_n - x_{n+1}$ to see whether it is always positive (in which case the sequence is decreasing) or always negative (in which case it is increasing) or sometimes positive and sometimes negative (in which case it cannot be monotonic).

2.2.2 Example

With

$$x_n = \frac{6n-1}{4n+3}$$

decide whether (x_n) is or is not monotonic.

Solution

In this case,

$$x_n - x_{n+1} = \frac{6n-1}{4n+3} - \frac{6(n+1)-1}{4(n+1)+3} = \frac{6n-1}{4n+3} - \frac{6n+5}{4n+7}$$

which comes to

$$\frac{-22}{(4n+3)(4n+7)}.$$

This is obviously negative for all n , so the sequence is increasing (and monotonic).

2.2.3 Example

With

$$x_n = 3 + \frac{4}{n} - \frac{2}{n^2}$$

show that (x_n) is a decreasing sequence.

Solution

In the present case,

$$x_n - x_{n+1} = 3 + \frac{4}{n} - \frac{2}{n^2} - \left(3 + \frac{4}{n+1} - \frac{2}{(n+1)^2} \right)$$

which simplifies to

$$\frac{4n^2-2}{n^2(n+1)^2}.$$

Since n is at least 1, this expression is always positive, so (x_n) is decreasing, as predicted.

2.2.4 Definition

A sequence is said to be *bounded* if the set of all its terms is bounded.

Likewise, a sequence is said to be *bounded above* if the set of all its terms is bounded above, and *bounded below* if the set of all its terms is bounded below.

2.2.5 Lemma

Every convergent sequence is bounded.

Proof

Suppose that $(x_n)_{n \in \mathbb{N}}$ converges and that its limit is L . By definition (and choosing $\varepsilon = 1$ for convenience) there is a positive integer n_0 such that all the terms of the sequence after the n_0^{th} lie between $L - 1$ and $L + 1$, that is, less than 1 unit distant from L . The earlier terms $x_1, x_2, x_3, \dots, x_{n_0-1}$ may well be further away from L , but there are only a finite number of them: so we can find the *biggest* distance from one of them to L ... call it M and now put $K = \max\{M, 1\}$. Then *every* x_n lies within the distance K from L , so $(x_n)_{n \in \mathbb{N}}$ is bounded.

2.2.6 Alert

The converse is not true! For instance, the “alternating sequence” $((-1)^n)_{n \in \mathbb{N}}$ is bounded (it lies entirely inside $[-1, 1]$, for example) but not convergent. However:

2.2.7 Theorem

1. A sequence that is increasing and bounded above must converge. (The limit is the supremum of the set of all its terms.)
2. A sequence that is decreasing and bounded below must converge. (The limit is the infimum of the set of all its terms.)

Proof

(1) Suppose that $(x_n)_{n \in \mathbb{N}}$ is increasing and bounded above. The set $X = \{x_n : n \in \mathbb{N}\}$ is non-empty and bounded above so, by completeness, its supremum (let's denote it by L) does exist. The definition of supremum then tells us that, if ε is any positive number:

- $x_n \leq L$ for every $n \geq 1$, and
- $L - \varepsilon < x_{n'}$ for at least one positive integer n' .

Now use the fact that the sequence is increasing, and we get:

$$L - \varepsilon < x_{n'} \leq x_{n'+1} \leq x_{n'+2} \leq x_{n'+3} \leq x_{n'+4} \leq \dots \leq L,$$

in other words, every term of the sequence from number n' onwards lies between $L - \varepsilon$ and L , whence $|x_n - L| < \varepsilon$ for every $n \geq n'$. Thus, $(x_n)_{n \in \mathbb{N}}$ converges to L .

2.2.8 Example

Find the infimum of the set

$$X = \left\{ 3 + \frac{4}{n} - \frac{2}{n^2} : n \in \mathbb{N} \right\}.$$

Solution

The *sequence* whose n^{th} term is the typical number in this set is decreasing and tends to 3. By the small print of the “bounded + monotonic” theorem, 3 has to be the infimum of the set!

2.2.9 Example

Let $t \in (0, 1)$ be a constant. Show that the sequence $(t^n)_{n \in \mathbb{N}}$ converges to 0.

Solution

Since $0 < t < 1$, all the powers of t are positive, and $t^n t < t^n 1 = t^n$, that is, $t^{n+1} < t^n$ for every n . So the given sequence is decreasing and bounded below by zero, and therefore must converge to *some* limit L . Now we need to identify L .

The sequence $(t^{n+1})_{n \in \mathbb{N}} = (t^2, t^3, t^4, t^5, \dots)$ is the original sequence with its first term removed so (see earlier comment) it also converges to (the same) L . Yet also (by part (4) of the algebra of limits) $t^{n+1} = t \cdot t^n$ has to converge to tL . That gives $L = tL$ (since limits are unique when they exist) which, since $t \neq 1$, forces $L = 0$ as predicted.

2.2.10 Theorem: limits across an inequality

If (a_n) and (b_n) are two convergent sequences such that $a_n \leq b_n$ for every $n \in \mathbb{N}$, then $\lim a_n \leq \lim b_n$.

Proof

For brevity, put $L_1 = \lim a_n, L_2 = \lim b_n$. If L_1 were not $\leq L_2$ then the number $\varepsilon = \frac{L_1 - L_2}{2}$ would be strictly greater than zero. Convergence tells us that, for sufficiently large n , both $|a_n - L_1|$ and $|b_n - L_2|$ will be smaller than ε , that is,

$$L_1 - \varepsilon < a_n < L_1 + \varepsilon, \quad L_2 - \varepsilon < b_n < L_2 + \varepsilon.$$

Our choice of ε , however, arranges that $L_1 - \varepsilon$ and $L_2 + \varepsilon$ are the same number! (Indeed, that is precisely why we chose it so.) Therefore (for large values of n like this) $b_n < a_n$, and this contradiction establishes the result.

Remarks

1. Be careful not to use this result on sequences about whose convergence you are unsure. For instance, $(-1)^n$ is certainly less than 2 for every value of n ... but does this tell us that $\lim(-1)^n \leq 2$? No, because $\lim(-1)^n$ does not exist.
2. Also be aware that the *strict* inequality $<$ is *not* preserved under limits in this way: that is, $a_n < b_n$ for all n does *not* guarantee that $\lim a_n < \lim b_n$. A simple illustration of this is that $-\frac{1}{n}$ is certainly strictly less than $+\frac{1}{n}$ for every n , and each tends to 0, but it would be foolish to claim that $0 < 0$ as a consequence.

2.2.11 Theorem: the “sandwich” or “squeeze”

Of three sequences $(a_n), (b_n), (c_n)$ suppose we know that $a_n \leq b_n \leq c_n$ for all n , and also that $(a_n), (c_n)$ converge to the same limit L . Then also $b_n \rightarrow L$.

Proof

Given any $\varepsilon > 0$ we can first use the given convergence to find positive integers N_a, N_c such that $|a_n - L| < \varepsilon$ for $n \geq N_a$ and $|c_n - L| < \varepsilon$ for $n \geq N_c$. Then for each $n \geq \max\{N_a, N_c\}$ we shall have both of these inequalities true at once, and so

$$L - \varepsilon < a_n < L + \varepsilon, L - \varepsilon < c_n < L + \varepsilon.$$

Combining pieces of this display with the given inequality $a_n \leq b_n \leq c_n$ we see that:

$$L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon$$

(for all $n \geq \max\{N_a, N_c\}$), which places b_n between $L \pm \varepsilon$, as required.

2.2.12 Example

Find the limit of the sequence whose n^{th} term is

$$\frac{6n - 5 \sin(n^2 + \pi\sqrt{n})}{2n + 3}.$$

Solution

The awkward-looking trig term varies between -5 and $+5$, so the n^{th} term here lies between $\frac{6n - 5}{2n + 3}$ and $\frac{6n + 5}{2n + 3}$. Since each of these converges to 3 via the algebra of limits, so must the given sequence.

2.3 Recursively defined sequences

Up to this point, every individual sequence that we have worked on has been specified by writing down either a formula for its n^{th} term or a list whose pattern was obvious enough that we could find such a formula easily. Not all important sequences work like this: for instance, although the pattern in the Fibonacci sequence $1, 1, 2, 3, 5, 8, 13, 21, 34, \dots$ is clear enough, it is far from obvious how to obtain an explicit formula for its n^{th} term; rather, the pattern is expressed by noting that each term from the third onwards is the total of the preceding two:

$$f_n = f_{n-1} + f_{n-2} \quad \text{provided that } n \geq 3$$

and, to complete the definition, $f_1 = 1, f_2 = 1$. This is an instance of what is called *recursive definition*.

Another instance that you are likely to have met is the Newton-Raphson approximation process. In this, faced with an equation of the form $f(x) = 0$ that we cannot solve precisely, but in which the function f can be differentiated we make an initial rough guess $x = x_1$ at the solution, perhaps via a sketch graph, and then improve that guess to a second one: $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$. In general, x_2 will be significantly closer to the true solution than x_1 was, and we can now repeat the improvement process

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}, \quad x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}, \quad x_5 = x_4 - \frac{f(x_4)}{f'(x_4)},$$

and so on. In most cases this sequence of improving approximations converges to a limit whose exact value is a solution to the original equation. The definition of this sequence

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)},$$

together with a random (but not grossly inaccurate) guess value for x_1 , is another example of recursive definition.

The essential characteristic of this style of definition is that, instead of telling us explicitly what the n^{th} term is, it provides us with an algorithm for determining it from the value(s) of one or more previous terms in the list. In time, that means that we could in principle work out any particular term that was of interest to us... but how could we hope to find the limit of such a sequence if, indeed, it had one?

2.3.1 Example

Identify, if it exists, the limit of the sequence $(x_n)_{n \geq 1}$ defined recursively by:

$$x_1 = 12; \quad x_{n+1} = \sqrt{3x_n + 28} \quad (n \geq 1).$$

Roughwork

With so little given information, we need to start by calculating the first few terms. They work out (to four decimal places) as follows:

$$12, 8, 7.2111, 7.0451, 7.0097, 7.0021, 7.0004, \dots$$

It must be stressed that this is very little evidence as to what happens as n goes to infinity! Nevertheless, it is enough to let us make a clutch of informed guesses: we guess that all the terms lie between 12 and 7, that the sequence is decreasing, and that the limit is 7. Now we have a definite proposal to try to establish:

Solution

- Step 0: For each $n \geq 1$ let $S(n)$ be the statement: $7 < x_n \leq 12$.
- Step 1: $S(1)$ says that $7 < 12 \leq 12$, which is true.
- Step 2: Assume the truth of a particular $S(k)$; that is, that $7 < x_k \leq 12$.
- Step 3: (How are we to get from x_k to the next term? As the recursive definition told us, we multiply by 3, add 28, and take the square root. So:)

$$21 < 3x_k \leq 36, \quad 49 < 3x_k + 28 \leq 64, \quad 7 < \sqrt{3x_k + 28} = x_{k+1} \leq 8 \leq 12.$$

In other words, $S(k+1)$ is also true.

By induction, all of the statements $S(n)$ are true, that is, all terms of the sequence do lie between 7 exclusive and 12 inclusive.

(Next, to compare x_n with x_{n+1} , the square root makes it less easy to see what is going on, so let us instead square both sides and actually compare x_n^2 with x_{n+1}^2 .)

Notice that

$$x_n^2 - x_{n+1}^2 = x_n^2 - 3x_n - 28 = (x_n - 7)(x_n + 4)$$

and because we already know that all terms lie between 7 and 12, that is a product of two *positive* numbers, therefore positive itself. So $x_n^2 > x_{n+1}^2$ and, taking (positive!) square roots, we get $x_n > x_{n+1}$ for all values of n , that is, the sequence is decreasing.

We now know that (x_n) is both bounded and decreasing, and must therefore converge to *some* limit L . Also (x_{n+1}) , being merely the sequence (x_n) without its first term, converges to the same L , and (x_n^2) converges to L^2 by algebra of limits. Take limits across the equation

$$x_{n+1}^2 - 3x_n - 28 = 0 \text{ for all } n$$

and we get the quadratic $L^2 - 3L - 28 = 0$ which factorises predictably into $(L - 7)(L + 4) = 0$. Thus, the only conceivable values of L are 7 and -4 . Yet $7 < x_n \leq 12$ (for all n) informs us that $7 \leq L \leq 12$, so -4 is really not a possible value, and only 7 remains. We conclude that (x_n) must indeed converge to the limit 7.

We'll do a second example, but this time cutting back the running commentary to the level that is normally expected in a solution.

2.3.2 Example

For the sequence (a_n) specified by the formulae $a_1 = 0, a_{n+1} = \sqrt{4a_n + 77}$:

1. Show that $0 \leq a_n < 11$ for every $n \geq 1$,
2. Show that the sequence is increasing,
3. Explain why it must possess a limit, and evaluate that limit.

Solution

- Step 0: For each n let $S(n)$ be the statement: $0 \leq a_n < 11$.
- Step 1: $S(1)$ says $0 \leq 0 < 11$ which is true.
- Step 2: Assume that $0 \leq a_k < 11$ for some particular k .
- Step 3: Then $4(0) + 77 \leq 4a_k + 77 < 4(11) + 77$, and so $0 \leq \sqrt{77} \leq \sqrt{4a_k + 77} = a_{k+1} < \sqrt{121} = 11$. Therefore $S(k + 1)$ is also true.

By induction, all the statements $S(n)$ are true. This proves (1).

Using (1), $a_n^2 - a_{n+1}^2 = a_n^2 - 4a_n - 77 = (a_n + 7)(a_n - 11)$ is the product of a positive and a negative, therefore negative. That is, $a_n^2 < a_{n+1}^2$ and so $a_n < a_{n+1}$ for each n . This proves (2).

Since (a_n) is now bounded and increasing, it must converge. Let L be its limit. Also $a_{n+1} \rightarrow L$. Taking limits across $a_{n+1}^2 - 4a_n - 77 = 0$ (for all n) gives $L^2 - 4L - 77 = (L + 7)(L - 11) = 0$, so L can only be -7 or 11. Yet $L = -7$ is impossible because every term is at least 0. Therefore $a_n \rightarrow 11$.

Chapter 3

Sampling a sequence: subsequences

3.1 Subsequences

Informally, a subsequence of a sequence is an endless list of its terms in their original order. So if $(x_n)_{n \geq 1}$ is any sequence, and we imagine its terms strung out in an unending list

$$x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10} \cdots$$

then a subsequence will be created by scanning along that list and lifting out an unending selection of what we find; for instance,

$$x_7, x_{16}, x_{21}, x_{22}, x_{39}, x_{122} \cdots$$

or

$$x_4, x_8, x_{13}, x_{400}, x_{401}, x_{605}, x_{677}, x_{759} \cdots$$

How should we best write a general subsequence when we don't know in advance *which* terms are to be picked out? We *could* relabel the first item chosen as y_1 and the second chosen as y_2 and the third as y_3 and so on, and the resulting symbol $(y_n)_{n \in \mathbb{N}}$ is certainly a perfectly good notation for a sequence, but it has lost any visible connection with the original sequence that we started with. A better method is to give labels to the *places* in the original sequence where we find the subsequence's items: if our first choice occurred at the n_1^{th} place in the original list, and our second at the n_2^{th} place, and our third at the n_3^{th} place and so on, then the chosen numbers that build up the subsequence are

$$x_{n_1}, x_{n_2}, x_{n_3}, x_{n_4} \cdots$$

and the entire subsequence can now be written as $(x_{n_k})_{k \geq 1}$. This is a slightly cluttered symbol, but it succeeds in capturing two of the important aspects of a subsequence: that it actually is an infinite sequence, and that its terms are *some* of the terms from the original (x_n) . The third important aspect – that the order has to be the same as was in the original (no back-tracking allowed) – is captured by insisting that $n_1 < n_2 < n_3 < n_4 < \cdots$, in other words, that the sequence of labels $(n_1, n_2, n_3, n_4, \cdots)$ has to be strictly increasing.

With that discussion behind us, we can now write down concisely what a subsequence is (and how to denote it):

3.1.1 Definition

Suppose given a sequence $(x_n)_{n \geq 1}$. For any strictly increasing sequence $(n_k)_{k \geq 1}$ of positive integers – that is, $n_1 < n_2 < n_3 < \cdots < n_k < \cdots$ – the sequence $(x_{n_k})_{k \geq 1} = (x_{n_1}, x_{n_2}, x_{n_3}, \cdots, x_{n_k}, \cdots)$ is a subsequence of it.

Notice how we used different symbols, in the sequence and in the subsequence, for “the thing that is going to infinity”. Indeed, we must be careful not to use a symbol with two meanings at once anywhere in mathematics, and especially not when working with subsequences: if we had denoted our subsequence by such a notation as $(x_{n_n})_{n \geq 1}$ then confusion would be practically guaranteed. Of course there is no need to use the particular letter k that we chose here: $(x_{n_i})_{i \geq 1}$ or $(x_{n_p})_{p \geq 1}$ would have done equally well.

3.1.2 Examples

1. For any sequence $(x_n)_{n \geq 1}$, the following are a few of its subsequences: the sequence of even-numbered terms $(x_{2k})_{k \geq 1}$, the sequence of odd-numbered terms $(x_{2k-1})_{k \geq 1}$, the sequence $(x_{k^2})_{k \geq 1} = (x_1, x_4, x_9, x_{16}, x_{25}, \cdots)$, the sequence $(x_{k!})_{k \geq 1} = (x_1, x_2, x_6, x_{24}, x_{120}, \cdots)$, the sequence $(x_{2^k})_{k \geq 1} = (x_2, x_4, x_8, x_{16}, x_{32}, \cdots)$.

2. In particular, $(5k-1)$ is a subsequence of (n) , (k^{-2}) is a subsequence of (n^{-1}) , $\left(\frac{1}{k!}\right)$ is a subsequence of $\left(\frac{1}{n}\right)$, $(\sin(7k^2 + 3k + 6))$ is a subsequence of $(\sin n)$, $(\sqrt[3k]{3k})$ is a subsequence of $(\sqrt[n]{n})$.

3. The sequence $\left((-1)^n \left(2 + \frac{1}{n}\right)\right)_{n \in \mathbb{N}} = \left(-2 - \frac{1}{1}, 2 + \frac{1}{2}, -2 - \frac{1}{3}, 2 + \frac{1}{4}, \cdots\right)$ does not converge, but two of its subsequences (those consisting of the odd terms and the even terms) are

$$\begin{aligned} & \left(-2 - \frac{1}{1}, -2 - \frac{1}{3}, -2 - \frac{1}{5}, -2 - \frac{1}{7}, \cdots\right) \\ &= \left((-1)^{2k-1} \left(2 + \frac{1}{2k-1}\right)\right)_{k \in \mathbb{N}} = \left(-\left(2 + \frac{1}{2k-1}\right)\right)_{k \in \mathbb{N}} \end{aligned}$$

and

$$\begin{aligned} & \left(2 + \frac{1}{2}, 2 + \frac{1}{4}, 2 + \frac{1}{6}, 2 + \frac{1}{8}, \cdots\right) \\ &= \left((-1)^{2k} \left(2 + \frac{1}{2k}\right)\right)_{k \in \mathbb{N}} = \left(2 + \frac{1}{2k}\right)_{k \in \mathbb{N}} \end{aligned}$$

which both do converge (to -2 and to 2 respectively).

4. The sequence $(n(1 + (-1)^n))_{n \in \mathbb{N}} = (0, 4, 0, 8, 0, 12, 0, 16, \cdots)$ is unbounded, but at least one of its subsequences (that consisting of the odd terms) is not only bounded but convergent; indeed, it is constant at 0. Another of its subsequences is $(2k(1 + (-1)^{2k}))_{k \in \mathbb{N}} = (4k)_{k \in \mathbb{N}} = (4, 8, 12, 16, \cdots)$ which is unbounded.
5. The sequence $(\sin(n\pi/2))$ is the “endlessly recycling” sequence

$$(1, 0, -1, 0, 1, 0, -1, 0, 1, \cdots)$$

from which we could evidently extract a subsequence converging to 1, another converging to -1 and a third converging to 0 (as well as many others that do not converge).

3.1.3 Theorem

Each subsequence of a convergent sequence converges, and to the same limit as does the original sequence.

Proof

Suppose that $x_n \rightarrow L$ and that $(x_{n_k})_{k \geq 1}$ is a subsequence of $(x_n)_{n \geq 1}$. We need to show that $x_{n_k} \rightarrow L$ (as $k \rightarrow \infty$).

Notice that, because $n_1 < n_2 < n_3 < \cdots < n_k < \cdots$ we get

$$n_1 \geq 1, n_2 \geq 2, n_3 \geq 3, \dots$$

and, in general,

$$n_k \geq k.$$

If $\varepsilon > 0$ is given, $x_n \rightarrow L$ tells us that there is a positive integer n_0 such that $n \geq n_0$ forces $|x_n - L|$ to be $< \varepsilon$. Now notice that $k \geq n_0$ forces $n_k \geq k \geq n_0$, that is, $n_k \geq n_0$, from which we get $|x_{n_k} - L| < \varepsilon$. (Less formally, big enough values of k guarantee that the error $|x_{n_k} - L|$ will be smaller than ε .) Hence $x_{n_k} \rightarrow L$, as required.

3.1.4 Example

To show that the sequence $\left((-1)^n \left(2 + \frac{1}{n}\right)\right)_{n \in \mathbb{N}}$ is divergent.

Solution

A page ago we discovered that this sequence has a subsequence whose limit is -2 and another whose limit is 2 . By the theorem, this could not happen if the full sequence converged. Hence the result.

3.2 Bolzano-Weierstrass: the overcrowded interval

There is an ancient mathematical insight sometimes called the pigeonhole principle: if you put $n + 1$ pigeons into n pigeonholes, then at least one pigeonhole will contain more than one pigeon. (More broadly, if you put $kn + 1$ or more letters into n mailboxes, where k and n are positive integers, then at least one mailbox must contain $k + 1$ or more letters.) The Bolzano-Weierstrass result starts out as an infinite version of this: if you distribute an infinite number of pigeons across two pigeonholes, then at least one pigeonhole will end up containing an infinite number of pigeons.

Let's re-express that in terms of sequences. If all the terms of a sequence lie within the union $A \cup B$ of two sets A, B of real numbers, then at least one of the two sets must include an infinite number of those terms. Formally: if $(x_n)_{n \in \mathbb{N}}$ is a sequence of elements

of $A \cup B$, then either A or B (or both) will include the number x_n for infinitely many values of n .

Hopefully, that sounds like little more than common sense... and yet it has an unobvious and powerful consequence:

3.2.1 The Bolzano-Weierstrass theorem

Every bounded sequence has a convergent subsequence.

Proof

Let $(x_n)_{n \in \mathbb{N}}$ be any bounded sequence. By its boundedness, we can find a closed interval $I_0 = [-M, M]$ (for some positive M) that includes *every* term x_n .

Now I_0 is the union of its left half and its right half: indeed, we can write $I_0 = [-M, 0] \cup [0, M]$. So we can pick one of the two halves that includes x_n for infinitely many values of n . Call whichever half we pick I_1 , and select also n_1 for which $x_{n_1} \in I_1$.

Next, repeat this argument upon I_1 : for I_1 is the union of its left half and its right half, and we can pick one of the two halves that includes x_n for infinitely many values of n . Call whichever half we pick this time I_2 , and select also n_2 **greater than** n_1 for which $x_{n_2} \in I_2$. The phrase **greater than** n_1 is legitimate because we have an infinite number of possible n_2 's to pick from, and can therefore surely arrange to make that choice larger than our previous n_1 , whatever it was.

Next, repeat this argument upon I_2 : I_2 is the union of its left half and its right half, and we can pick one of the two halves that includes x_n for infinitely many values of n . Call whichever half we pick this time I_3 , and select also n_3 **greater than** n_2 for which $x_{n_3} \in I_3$.

And so on without end (appeal to induction if you feel that it's necessary).

This generates a sequence $(I_1, I_2, I_3, \dots, I_k, \dots)$ of closed intervals, each contained in the previous one and exactly half of its length, and also a subsequence $(x_{n_1}, x_{n_2}, x_{n_3}, \dots, x_{n_k}, \dots)$ of the original sequence, such that (for all k) x_{n_k} belongs to I_k . All we still need to do is to use the "shrinking" nature of the intervals to show that the subsequence converges.

Write the typical I_k as $[a_k, b_k]$ and notice that, since the length of I_1 was M and the (half) length of I_2 was $M/2$ and the length of I_3 was $M/4$ and so on, b_k is actually $a_k + M/2^{k-1}$. Since I_{k+1} is either the left or the right half of I_k , we also have

$$a_k \leq a_{k+1} \leq b_{k+1} \leq M$$

for all k , so (a_k) is an increasing bounded sequence, and therefore converges to some limit L . Finally,

$$a_k \leq x_{n_k} \leq b_k = a_k + M/2^{k-1}$$

lets us appeal to the squeeze because a_k and $a_k + M/2^{k-1}$ both converge to (the same) L . Thus we have (x_{n_k}) converging to L .

For the purposes of this text, the really important applications of Bolzano-Weierstrass happen only after we have defined *continuous functions*.